

Contact

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Education

Universiti Teknologi Malaysia
Bachelor of Artificial Intelligence with Honours
2024 - 2027

Kolej Matrikulasi Selangor
Physical Science
2023 - 2024
CGPA: 4.00

Skills

Programming

- Python
- C++
- HTML
- SQL

Languages

- English (Fluent)
- Malay (Fluent)
- Tamil (Proficient)

Certification

- Fundamentals of Cloud Computing – RunCloud
- Google IT Automation with Python Professional Certificate
- Google IT Support Professional Certificate
- IBM Data Analyst Professional Certificate
- GCI World April 2026 - Data Science Certificate
- Artificial Intelligence Fundamentals - IBM SkillBuild

NIVEESHANA BALACHANDRAN

AI Undergraduate

Artificial Intelligence undergraduate at Universiti Teknologi Malaysia (UTM) with a strong foundation in machine learning, data science, programming, and AI-driven solutions. Experienced in developing academic and practical projects involving computer vision, predictive modelling, reinforcement learning, IoT, and automation. Skilled in Python, machine learning frameworks, data analysis, and problem-solving. Demonstrated leadership experience through university programme management and STEM outreach initiatives. Seeking an internship opportunity to apply technical knowledge, gain industry experience, and contribute to practical AI solutions

LEADERSHIP & PROFESSIONAL ENGAGEMENT

President, Persatuan Mahasiswa Hindu (PMH), UTM KL | 2025–2026

- Led the society in planning and executing university programmes.
- Managed documentation, approvals, logistics, and stakeholder coordination.
- Led STEM outreach programmes in Robotics, Engineering, and Programming.
- Coordinated teams and delegated responsibilities for effective programme execution.

VYAAB 2026 – United Malaysian Hindu Businesses Summit | PMH Collaborator

- Represented PMH UTM KL in its collaboration with the 1st United Malaysian Hindu Businesses Summit.
- Supported cross-sector engagement connecting students, entrepreneurs, professionals, and business leaders.

INNOTECH 3.0 – Future Innovation Summit | PMH Co-organizer

- Led PMH UTM KL's involvement as an event partner during my presidency.
- Supported youth engagement in Artificial Intelligence, Responsible AI, and emerging technologies through institutional collaboration.

STEM & COMMUNITY PROJECT

Puthiya Manitha – STEM Outreach | Programme Director

- Conducted hands-on Programming, Robotics, and Engineering activities for 7 primary school.
- Coordinated facilitators, equipment, school communication, and programme execution.

Robomania – Robotics Workshop | Deputy Director

- Organised a robotics workshop for approximately 90 students.
- Coordinated facilitators, equipment, and hands-on robotics activities.

RoboXplore – Robotics Workshop | Deputy Director

- Co-organised a STEM robotics programme for 30 secondary school students.
- Coordinated facilitators, logistics, technical activities, and programme execution.
- Facilitated programming and robotics activities to promote STEM learning.

Projects

RainHarvest – Smart Rainwater Harvesting & Automated Irrigation System

- Developed an ESP32-S3 IoT system for rainwater monitoring and automated irrigation.
- Integrated sensors, OLED display, water pump, and Telegram remote control.
- Explored n8n for intelligent Telegram automation.

Smart Flow Rover - ESP32, C++, IoT

- Developed an ESP32-based drainage monitoring rover with ultrasonic, gas, and IR sensors for blockage and environmental monitoring.
- Implemented DC motor control and PWM-based movement using L298N and PlatformIO.
- Designed a 7.4V battery and regulated power system, with planned MQTT/HiveMQ and cloud monitoring integration.

SENTINEL RFID Monitoring System

- Developed an RFID monitoring system using ESP32 and UHF RFID communication.
- Implemented RS485 communication for RFID data transmission.
- Integrated Cloudflare Workers and Supabase for cloud-based RFID event logging.
- Designed logic to reduce duplicate or unwanted RFID readings.

Tools & Technologies

- Git & GitHub
- VS Code
- Jupyter Notebook
- PlatformIO
- Arduino / ESP32
- TensorFlow.js
- n8n
- Supabase
- Cloudflare Workers

Professional Skills

- Project Management
- Leadership
- Teamwork
- Communication
- Problem Solving
- Critical Thinking
- Event Coordination

Smart Parking System | IoT & Embedded Systems

- Developed a smart parking solution to monitor parking space occupancy and availability through automated detection.
- Integrated sensor-based monitoring to support efficient parking management and real-time status updates.

ChainGuardian – Autonomous Supply Chain Disruption Resolver

- Collaborated with Team NVK CODERS to develop an autonomous AI-powered supply chain disruption resolver using Google Gemini 2.0 Flash.
- Implemented agentic workflows for disruption analysis, ERP querying, inventory checking, alternative supplier identification, and automated resolution email generation.
- Integrated Node.js, Express, Socket.IO, and Supabase for backend services, real-time communication, and database management.

Optify – Glaucoma Image Classification

- Developed a deep learning-based solution for glaucoma image classification using TensorFlow/Keras.
- Applied image preprocessing and explored TensorFlow.js for model deployment.

ACHIEVEMENTS

First Runner-Up – Cisco AI Hackathon

- Secured First Runner-Up by collaborating with a team to develop an AI-powered educational game prototype.
- Designed an interactive learning solution that evaluates students' performance through an integrated AI application.

Dean's Award – Universiti Teknologi Malaysia (UTM)

- Recognised with the Dean's Award for academic excellence in the Bachelor of Artificial Intelligence programme.
- Demonstrated consistent academic performance, dedication, and commitment to learning.

References

Dr. Pritheega Malalingam
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